guarantee it the ability to read data at a specific rate. If this assurance is possible, the system ensures that the application gets the required access to the I/O systems. This also creates a reservation on the I/O such that the system won't service other applications until the needs of the multimedia application have been met.

Previous versions of Windows have limited the transaction unit for I/O operations to 64 KB. This means that if an application need to deliver data larger than 64 KB to a disk, it needs to make multiple trips to the I/O system. This puts unnecessary load on the systems involved and adds unneeded overhead for transactions. Under Vista, therefore, this upper limit has been removed, and storage I/O request sizes are no longer capped. Explorer's Copy command, for example, now issues 1 MB data transactions.

3.6 Memory Management

A 32-bit operating system is limited to addressing 4 GB of memory space. The Windows kernel however, can only access half of that space by default; the other 2 GB is allocated to other threads running on the processor. Within the limited 2 GB, the kernel is require to shuffle various bits and pieces of itself which are vital to the efficient running of the OS—details such as device drivers, file system cache, buffers, and so on.

Prior to Vista, this 2 GB address space was allocated at boot time. Such a preset allocation led to inefficient use of the limited space as some areas required more space post-boot while others were assigned too much of the 2 GB. This, of course, led to application failures and drivers from completing I/O operations.

The 64-bit version of Vista can address far more memory space than the 32-bit counterpart—this number is so high that the practical ceiling will not be reached for a good time to come. The 32-bit version of Vista does face the same ceiling, however. To address